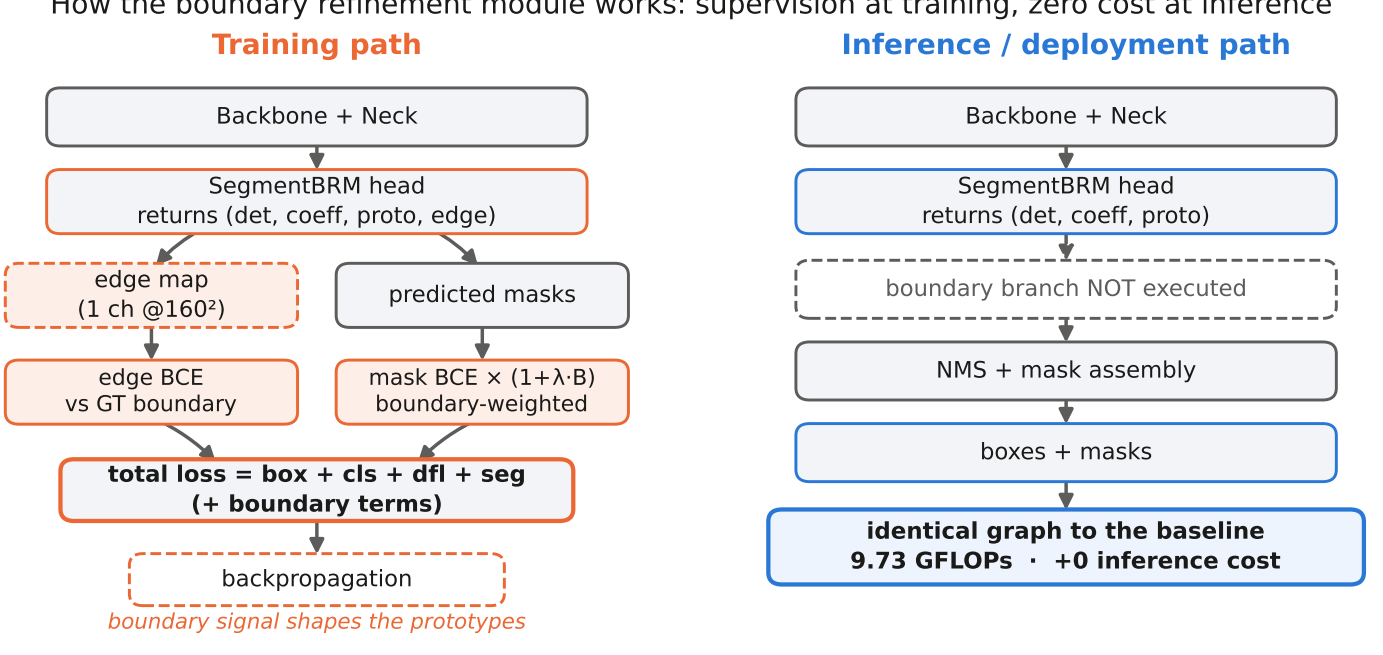
How the boundary refinement module works: supervision at training, zero cost at inference
Training path
Backbone + Neck
SegmentBRM head
returns (det, coeff, proto, edge)
edge map
(1 ch @160²)
predicted masks
edge BCE
vs GT boundary
mask BCE × (1+λ·B)
boundary-weighted
total loss = box + cls + dfl + seg
(+ boundary terms)
backpropagation
boundary signal shapes the prototypes
Inference / deployment path
Backbone + Neck
SegmentBRM head
returns (det, coeff, proto)
boundary branch NOT executed
NMS + mask assembly
boxes + masks
identical graph to the baseline
9.73 GFLOPs · +0 inference cost